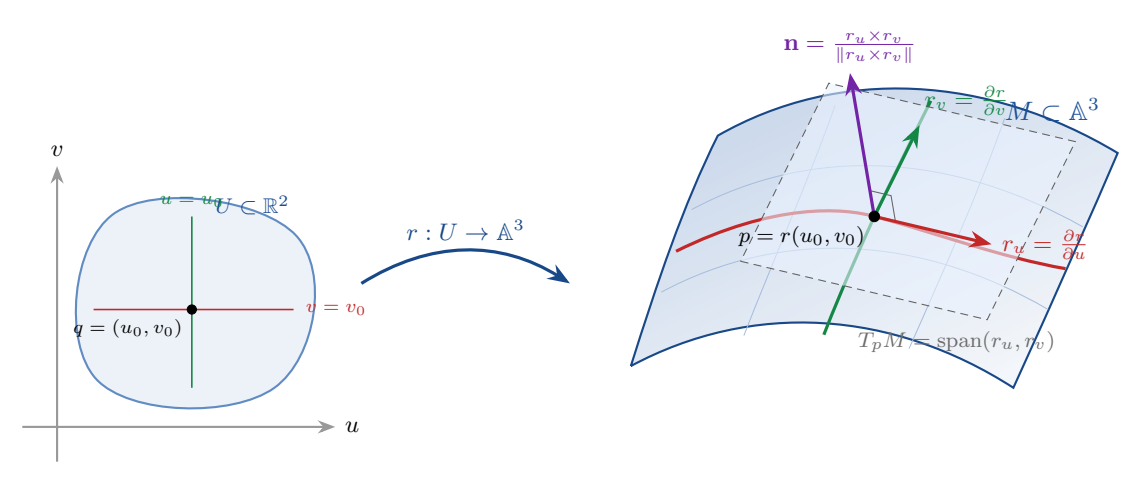

v
$u = u_0$
$U \subset \mathbb{R}^2$
$r : U \to \mathbb{A}^3$
$v = v_0$
$q = (u_0, v_0)$
u
$\mathbf{n} = \frac{r_u \times r_v}{\|r_u \times r_v\|}$
$r_v = \frac{\partial r}{\partial v}$
$M \subset \mathbb{A}^3$
$p = r(u_0, v_0)$
$r_u = \frac{\partial r}{\partial u}$
$T_pM = \mathrm{span}(r_u, r_v)$